AI-Powered Delivery Exception Resolution

Project 3 | Low-Code Multi-Agent System

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- Executive summary and verified outcomes
- Business problem and solution approach
- Data loading, model initialization, and retrieval
- Tools, multi-agent architecture, escalation, and security
- Test cases, traces, document citations, and evaluation metrics
- Conclusions, recommendations, and appendix

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Executive Summary

The completed notebook was executed against all attached data.

SCOPE
13 raw delivery-log rows
10 unique shipments
12 customers + 6 lockers
10-page operations playbook

RESULT
Task completion: 100%
Escalation accuracy: 100% (8/8)
Tool-call accuracy: 100%
Coherence: 5.00 / 5 (offline trace checklist)

RECOMMENDATION
Keep deterministic policy rules authoritative for safety and escalation. Use the LLM for classification, rationale, and customer communication.

- The system filters routine noise before unnecessary tool or model calls.
- It removes duplicate scans, retrieves policy context, checks lockers, and applies escalation rules.
- Structured Resolution and Communication agents are validated by critic agents before finalization.
- The run used a credential-free deterministic fallback; remote GPT-4o-mini mode remains supported when configured.

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Business Problem Overview and Solution Approach

BUSINESS PROBLEM

Delivery exceptions are messy, repetitive, and policy-sensitive. Duplicate scans create false work; natural-language notes hide the relevant facts; inconsistent escalation risks customer and operational harm.

SOLUTION

A hub-and-spoke LangGraph workflow combines guardrails, deterministic tools, structured agent outputs, critic validation, and traceable final actions.

- 1. Preprocess: deduplicate, consolidate attempts, scan for injection, and filter routine noise.
- 2. Enrich: retrieve customer profile, locker availability, playbook chunks, and escalation signals.
- 3. Decide: Resolution Agent selects classification and action with a policy-grounded rationale.
- 4. Validate: Resolution Critic may accept, revise, or escalate; Communication Critic validates the message.
- 5. Finalize: package an auditable action, escalation flag, customer message, citations, and latency.

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Data Loading

Source coverage and attributes used by the solution

DELIVERY LOGS | 13 ROWS

shipment_id, timestamp, status_code, driver description, customer_id, address, package type/size, attempt number, duplicate flag

CUSTOMER DB | 12 ROWS

Customer tier, preferred channel, exception count in the last 90 days, active credit, and name. Names are redacted by default.

LOCKER DB | 6 ROWS

Locker address and ZIP, capacity status, operating hours, and maximum package size. Eligibility is checked before reroute communication.

PLAYBOOK + LABELS

10-page PDF split into 24 page-aware chunks. Ground truth has 13 labels aligned to the raw logs and consolidates to 10 unique shipment scenarios.

- Important data quality handling: duplicate scans for SHP-002 and SHP-007 are removed before consolidation.
- Ground truth is consolidated by shipment, using the highest-attempt exception label for multi-row shipments.

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Model Initialization and Retrieval

REMOTE MODE
Generation: gpt-4o-mini
Validation: gpt-4o-mini
Temperature: 0.0
Credentials loaded only from config.json or environment

VALIDATION MODE
The executed submission used a deterministic offline fallback because no credentials were attached. This makes the evaluation reproducible and still exercises the complete graph.

RETRIEVAL
ChromaDB cosine search over 24 local TF-IDF vectors returns ranked policy chunks with source-page metadata.
No external embedding download is required.

- Pydantic schemas enforce valid Resolution, Communication, and Critic outputs.
- Temperature 0.0 is used to reduce variance in structured decisions.
- The notebook supports remote LLM execution when a valid config.json is supplied, but never hard-codes credentials.
- The offline path is explicitly labeled in the exported results as runtime_mode=offline_fallback.

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Defining Tools

Delivery Log Reader
Reads all raw CSV rows and feeds the preprocessor.

Customer Profile Lookup
Parameterized SQLite query; PII name is returned only with explicit include_pii=True.

Locker Availability Checker
Checks ZIP, size capacity, FULL/LIMITED status, and returns eligibility plus reason.

Playbook Search
Retrieves relevant page-aware policy chunks for the Resolution and Critic agents.

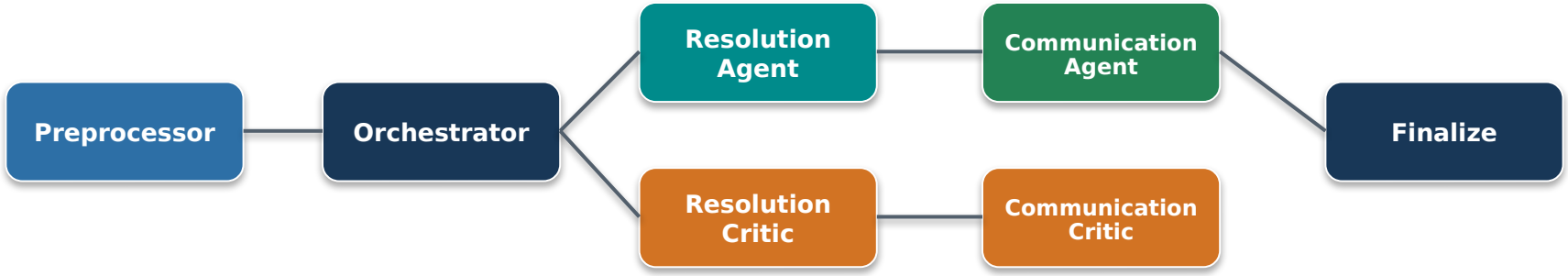
Escalation Rule Engine
Deterministically evaluates automatic and discretionary triggers, including delay-hour parsing.

- Evaluation result: 10/10 scenarios invoked the expected tool path, including the intended noise short-circuit.

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Multi-Agent System Architecture

Hub-and-spoke orchestration keeps routing deterministic while allowing agent outputs to be structured and critic-checked.



- Prompts: policy-grounded resolution, tier-aware communication, resolution validation, communication validation, and 1–5 trajectory coherence judging.
- Escalation: automatic triggers (third attempt, VIP ≥3 exceptions, damaged perishables, suspected fraud, safety) plus discretionary triggers (high-history Standard and delayed Premium perishables).
- Security: parameterized SQL, default PII redaction, prompt-injection scanning before LLM calls, no raw PII in coherence judging, and safe fallback escalation.

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Test Cases Execution, Evaluation, and Observability

AGGREGATE METRICS
Task completion: 100%
Escalation accuracy: 100% (8/8)
Tool-call accuracy: 100%
Coherence: 5.00 / 5 (offline checklist)
Offline latency: 0.01s (not a production benchmark)

OBSERVABILITY
Every exception trace records tool calls, agent decisions, critic decisions, escalation signals, latency, and retrieved playbook page citations.

ACCURACY COMMENT
All expected resolutions, tones, exception classifications, and escalation decisions match the supplied ground truth for the 10 unique scenarios.

- SHP-001 Delivered → noise / no action
- SHP-002 VIP attempts → reschedule / escalate
- SHP-003 Address issue → reschedule
- SHP-004 VIP fragile damage → replace / escalate
- SHP-005 Third attempt → locker reroute / escalate
- SHP-006 Refused → return to sender
- SHP-007 Perishable weather >4h → replace / escalate
- SHP-008 Standard high history → reschedule / escalate
- SHP-009 Damaged perishable → replace / escalate
- SHP-010 Depot scan → noise / no action

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APPENDIX

Detailed scenario outcomes, citations, and submission checklist

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Appendix: Detailed Evaluation Results

Shipment	Outcome / tone	Esc.	Task	Tools	Coherence	Trace / observation	Playbook
SHP-001	NO EXCEPTION / —	N/A	PASS	PASS	5/5	Preprocessor → delivered-noise gate → finalize.	N/A
SHP-002	RESCHEDULE / formal	PASS	PASS	PASS	5/5	Dedup → profile/RAG → reschedule → critics → escalate.	P2, 5, 6
SHP-003	RESCHEDULE / casual	PASS	PASS	PASS	5/5	Profile/RAG → address retry → both critics accept.	P3, 2, 3
SHP-004	REPLACE / formal	PASS	PASS	PASS	5/5	Profile/RAG → replace → critics → escalate.	P4, 4, 7
SHP-005	LOCKER REROUTE / casual	PASS	PASS	PASS	5/5	Locker/RAG → reroute → critics → escalate.	P2, 2, 3
SHP-006	RETURN / casual	PASS	PASS	PASS	5/5	Profile/RAG → return to sender → critics accept.	P5, 5, 4
SHP-007	REPLACE / formal	PASS	PASS	PASS	5/5	Weather/perishable RAG → replace → escalate.	P6, 2
SHP-008	RESCHEDULE / casual	PASS	PASS	PASS	5/5	History rule → reschedule → discretionary escalation.	P2, 5, 5
SHP-009	REPLACE / formal	PASS	PASS	PASS	5/5	Damage/perishable RAG → replace → escalate.	P4, 6, 2
SHP-010	NO EXCEPTION / —	N/A	PASS	PASS	5/5	Preprocessor → routine-scan gate → finalize.	N/A

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Conclusions and Business Recommendations

WHAT WORKED

All 10 unique shipment scenarios were completed correctly. Duplicate scans and noise were filtered, policy-based escalation was enforced, and customer tone matched tier expectations.

NEXT PRODUCTION STEPS

Persist shipment notes and supervisor queues, add temperature-controlled locker metadata, and expand adversarial tests for malformed records and API outages.

- Keep deterministic policy rules authoritative for safety and escalation.
- Use the LLM for flexible language tasks: classification rationale and customer-facing communication.
- Monitor task completion, escalation accuracy, tool-call accuracy, coherence, and latency in production.
- Keep the offline fallback as a reproducible evaluation and disaster-recovery path.

Submission package verified against attached ground truth

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